

MEHMET KEREM TURKCAN

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SUMMARY

Associate Research Scientist at Columbia University with 8+ years of experience building computer-vision, multimodal, and scientific data systems for environments where latency, reliability, and real-world constraints matter. I lead research from data and model design through optimization, deployment, evaluation, and public release, with work spanning urban infrastructure, medical robotics, and connectome-scale neuroscience. My record includes systems deployed across major urban testbeds, publications at CVPR, ICML, ACM UIST, ACM/IEEE SEC, IEEE INFOCOM, and eLife, and open-source projects with 1,200+ combined GitHub stars.

SELECTED CONTRIBUTIONS

- ♦ **Real-world AI systems:** Lead multimodal perception and urban-analytics efforts intended to scale across 900+ New York City intersections and the COSMOS PAWR, DataCity, and FAU MobIntel testbeds.
- ♦ **Frontier-model deployment:** Created DART, a real-time open-vocabulary detector with TensorRT deployment, public weights, and 300+ GitHub stars; led UrbanOmniDetect, an oral paper at the CVPR 2026 DriveX Workshop.
- ♦ **Research leadership:** Collaborate across an NSF Engineering Research Center spanning 5 institutions and 30+ principal investigators; received project support from NVIDIA and EmpireAI and challenge recognition from INRIX x MetroLab.
- ♦ **Open scientific platforms:** Built FlyBrainLab's TypeScript/JupyterLab interface, OrientDB-backed graph and ontology systems, large-scale query layer, retrieval and question-answering workflows, GPU simulation, and interactive 3D visualization.
- ♦ **Teaching and mentorship:** Taught Columbia graduate deep-learning courses, mentored Master's and high-school researchers, and served as the main engineering instructor for CS3's Research Experience for Teachers in 2024–2026.

EXPERIENCE

- ♦ **Columbia University**, New York, NY Jul 2024 – Present
Associate Research Scientist, Civil Engineering & Engineering Mechanics
 - Lead large-scale applied AI projects in real-time video analytics, open-vocabulary and 3D perception, edge/VLM systems, urban digital twins, and medical-robotics foundation models; manage shared high-performance computing and cross-team engineering.
 - Created DART, converting promptable frontier vision models into a real-time multi-class detector through shared-backbone inference, batched decoding, adapter distillation, and TensorRT FP16 optimization.
 - Led UrbanOmniDetect, a calibration-free, view-agnostic monocular 3D detector for ego-vehicle, infrastructure, and aerial cameras, with public models and the UrbanOmniView dataset.
 - Contributed to Open-H-Embodiment and surgical world-model research for healthcare robotics; developed public model assets for video generation, phase recognition, metric depth, and skills assessment.
 - Co-authored the ACM/IEEE SEC 2025 Best Paper on privacy-preserving urban safety analytics, ICML 2025 work on adaptive data collection, and 2026 papers at CVPR Workshops and IEEE INFOCOM.
 - Designed and led the engineering curriculum for three annual CS3 Research Experience for Teachers cohorts, translating current AI research into reproducible K–12 lesson plans and classroom projects.
- ♦ **Columbia University**, New York, NY Nov 2022 – Jul 2024
Postdoctoral Research Scientist
 - Built real-time digital twins of New York City intersections integrating object detection, multi-object tracking, trajectory forecasting, simulation, and safety-warning generation.
 - Developed synthetic-data and benchmarking systems for dense urban perception, including Boundless in Unreal Engine 5 and the Constellation high-altitude object-detection dataset.
 - Built healthcare AI pipelines with Northwell Health collaborators for surgical phase recognition and automated endoscopy skill scoring, resulting in peer-reviewed journal publications and clinical presentations.
 - Mentored 11 Master's researchers and three high-school researchers across computer vision, generative modeling, and urban analytics projects.
- ♦ **Columbia University**, New York, NY Jan 2023 – Dec 2023
Lecturer in Electrical Engineering
 - Designed and taught ECBM E4040 Neural Networks & Deep Learning and EECS E6691 Advanced Deep Learning, including lectures, coding assignments, exams, and conference-style final projects.
 - Co-taught VOA1 0003E Neural Networks and Deep Learning for the Columbia Video Network AI Executive Certificate in Fall 2022 and Summer 2023.

◇ **Columbia University**, New York, NY
Doctoral Researcher, Bionet Group

Jan 2017 – Oct 2022

- Co-created FlyBrainLab, an open computing platform for natural-language exploration, graph querying, simulation, and visualization of connectome-scale neuroscience data.
- Developed an early research-agent architecture combining programmable ontologies, literature-linked entity retrieval, dense passage retrieval, biomedical BERT question answering, graph-database tools, and executable scientific workflows.
- Built scalable TypeScript and Python interfaces, OrientDB/NeuroArch query services, GPU-backed neural simulations, and interactive 3D tools used in open neuroscience research and education.

◇ **Columbia University and ArtGe Technologies**

2014 – 2016

Research and R&D Roles

New York, NY and Istanbul

- Developed deep-learning systems for thermal-camera infrastructure inspection, solar microgrid control, intelligent-building forecasting, stadium-scale face recognition, and agricultural stereo vision.

EDUCATION

◇ **Columbia University**, New York, NY

2022

- Ph.D. in Electrical Engineering; Systems Biology and Neuroengineering, GPA 4.10.

◇ **Columbia University**, New York, NY

2016

- M.S. in Computer Science; Machine Learning and Thesis Track, GPA 3.97.

◇ **Istanbul Technical University**, Istanbul, Turkey

2015

- B.S. in Electronics and Communication Engineering, GPA 3.75.

AWARDS, GRANTS, AND RECOGNITION

◇ **Best Paper Award**, ACM/IEEE Symposium on Edge Computing

2025

◇ NVIDIA research grant, *Edge AI for Equitable and Safe Intersections in Urban Metropolises*

2025

◇ EmpireAI award, *AI Computing for Community, Health, Safety and Climate*

2025

◇ INRIX x MetroLab Challenge finalist

2025

◇ Columbia University Electrical Engineering Distinguished Teaching Award

2024

◇ Smart Cities North America Award winner in Transportation, IDC Government Insights

2023

◇ National Postdoc Appreciation Week Excellence Award, Columbia University

2023

◇ Herbert French Fellowship, Columbia University; Helmsley Fellowship, CSHL

2017–2019

SELECTED SOFTWARE AND DATA

◇ **DART**: Real-time open-vocabulary detection; PyTorch, TensorRT, distillation, public weights, and deployment benchmarks.

◇ **UrbanOmniDetect**: Calibration-free monocular 3D detection with public models and the UrbanOmniView multi-view dataset.

◇ **Loom**: Analytical looped-transformer computer with C compilation, ONNX/WebGPU demos, FPGA verification, and 140+ tests.

◇ **FlyBrainLab**: Natural-language and ontology interfaces, OrientDB graph querying, GPU simulation, and 3D scientific visualization.

◇ **Boundless**: Unreal Engine 5 synthetic-data pipeline and datasets for infrastructure and aerial object detection.

◇ **bikeped**: Real-time fisheye perception, safety alerts, simulator scenarios, and conformance testing for urban intersections.

◇ **GPTune**: Early GPT-2 fine-tuning and sampling toolkit for custom language-model adaptation.

SELECTED PUBLICATIONS AND PREPRINTS

1. **Calibration-Free View-Agnostic Monocular 3D Object Detection for Urban Scenes.** M. K. Turkcan, D. Gumaste, and Z. Kotic. *CVPR Workshops (DriveX)*, oral presentation, 2026.
2. **Worst-Case Attacks in Reactive Edge-Cloud Systems: Expected Latency and SLA Violation.** J. Tavori, M. K. Turkcan, Z. Kotic, J. Ghaderi, and G. Zussman. *IEEE INFOCOM*, 2026.
3. **Digital Eyes on the Road: Using Street Cameras to Verify Traffic Integrity and Detect Sybil Attacks at City Scale.** J. Tavori, M. K. Turkcan, Z. Kotic, S. Stolfo, and G. Zussman. *USENIX VehicleSec*, to appear, 2026.
4. **Open-H-Embodiment: A Large-Scale Dataset for Enabling Foundation Models in Medical Robotics.** N. Nelson, J.-T. Chen, J. Haworth, et al., including M. K. Turkcan. *arXiv:2604.21017*, 2026.
5. **Detect Anything in Real Time: From Single-Prompt Segmentation to Multi-Class Detection.** M. K. Turkcan. *arXiv:2603.11441*, 2026.
6. **Harnessing Floating Car Data, Traffic Camera Observations, and Network Flow Analysis for Traffic Volume Estimation.** A. Kosikova, M. K. Turkcan, A. Darrat, and A. Smyth. *arXiv:2605.09891*, 2026.

7. **Computer Vision Scoring of Endoscopically Traced Line Figures in an Inanimate Scope Tip Coordination Training Model.** N. Mitra, S. Narasimhan, **M. K. Turkcan**, et al. *Surgical Endoscopy*, 2026.
8. **Real-Time Video Analytics for Urban Safety: Deployment over Edge and End Devices.** M. Ghasemi, Y. Fu, X. Ouyang, P. Wang, **M. K. Turkcan**, et al. *ACM/IEEE SEC, Best Paper Award*, 2025.
9. **Adaptive Data Collection for Robust Learning Across Multiple Distributions.** C. Zang, **M. K. Turkcan**, G. Zussman, Z. Kostic, and J. Ghaderi. *ICML*, 2025.
10. **Distributed VLMs: Efficient Vision-Language Processing through Cloud-Edge Collaboration.** Y. Li, D. Gumaste, **M. K. Turkcan**, J. Ghaderi, G. Zussman, and Z. Kostic. *IEEE PerCom Workshops*, 2025.
11. **Towards Suturing World Models: Learning Predictive Models for Robotic Surgical Tasks.** **M. K. Turkcan**, M. Ballo, F. Filicori, and Z. Kostic. *arXiv:2503.12531*, 2025.
12. **StreetNav: Leveraging Street Cameras to Support Precise Outdoor Navigation for Blind Pedestrians.** G. Jain, B. Hindi, Z. Zhang, et al., including **M. K. Turkcan**. *ACM UIST*, 2024.
13. **Boundless: Generating Photorealistic Synthetic Data for Object Detection in Urban Streetscapes.** **M. K. Turkcan**, Y. Li, C. Zang, J. Ghaderi, G. Zussman, and Z. Kostic. *arXiv:2409.03022*, 2024.
14. **Additional first-author systems preprints in 2026:** Loom analytical neural computing, bike-pedestrian safety, and city-scale congestion-pricing analysis.
15. **A Programmable Ontology Encompassing the Functional Logic of the Drosophila Brain.** A. A. Lazar, **M. K. Turkcan**, and Y. Zhou. *Frontiers in Neuroinformatics*, 2022.
16. **FlyBrainLab: Accelerating Discovery of the Functional Logic of the Drosophila Brain.** A. A. Lazar, T. Liu, **M. K. Turkcan**, and Y. Zhou. *eLife*, 2021.

TEACHING AND CURRICULUM LEADERSHIP

- ◊ **Main Engineering Instructor, CS3 Research Experience for Teachers, 2024–2026:** Designed and led a four-week curriculum covering neural networks, model evaluation, computer vision, object tracking, generative AI, annotation, YOLO training, and edge deployment; supported teachers in producing classroom-ready lesson plans.
- ◊ **Lecturer, Columbia University:** EECS E6691 Advanced Deep Learning, Spring 2023; ECBM E4040 Neural Networks & Deep Learning, Fall 2023; VOAI 0003E Neural Networks and Deep Learning, Fall 2022 and Summer 2023.
- ◊ **Course and teaching support:** Computing with Brain Circuits of Model Organisms, 2020–2021; Fruit Fly Brain as a Neuroinformation Processor, 2018; Computational Neuroscience, 2017.

SELECTED PRESENTATIONS

- ◊ **CS3 NSF Annual Site Visit:** DART and UrbanOmniDetect demonstrations, May 12–15, 2026.
- ◊ **From Sensors to Systems: Real-Time Traffic Analysis for Faster Decision-Making**, panelist, CS3 Innovation Summit, Feb. 5, 2026.
- ◊ **Port Authority Tomorrow Summit**, workshop host, World Trade Center, Jan. 7, 2026.
- ◊ **Evaluating Micromobility & Dangerous Riding Behaviors**, Vision Zero Research on the Road, Nov. 19, 2025.
- ◊ **FlyBrainLab: An Interactive Open Computing Platform**, PyData Global, Nov. 2020.

PROFESSIONAL SERVICE

Peer reviewer: CVPR 2026; ICML 2026; ICMLA 2026; WACV 2025; ACM MobiCom 2024; IEEE/ACM ToN; IEEE TVLSI; Sensors, Electronics, and Smart Cities (MDPI).

TECHNICAL SKILLS

- ◊ **AI and deployment:** PyTorch, JAX, TensorFlow, Hugging Face, TensorRT, ONNX, WebGPU, NVIDIA Jetson; detection, tracking, 3D perception, VLM/VLA systems, RAG, agents, diffusion, and world models.
- ◊ **Languages, systems, and data:** Python, C/C++, CUDA, TypeScript/JavaScript, SQL, MATLAB; OrientDB, RPC, distributed querying, HPC, Docker, GitHub Actions; large-scale data collection and annotation, synthetic data, Unreal Engine 5, JupyterLab, Three.js/WebGL, scientific visualization, and Adobe Creative Suite.